



Revolution in software distribution called Docker

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Y Soft: Technology Hour Prague, 25th Sep 2014

Before we start...

Did you upgrade bash yet?

```
$ env x='() { :;; echo vulnerable' bash -c "echo this is a test"
vulnerable
this is a test
```

- <https://access.redhat.com/articles/1200223>
- <https://access.redhat.com/security/cve/CVE-2014-6271>
- <https://access.redhat.com/security/cve/CVE-2014-7169>

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- JBoss Software Engineer
 - Cloud-related stuff
 - Docker
- Fedora packager
 - `yum install wildfly`

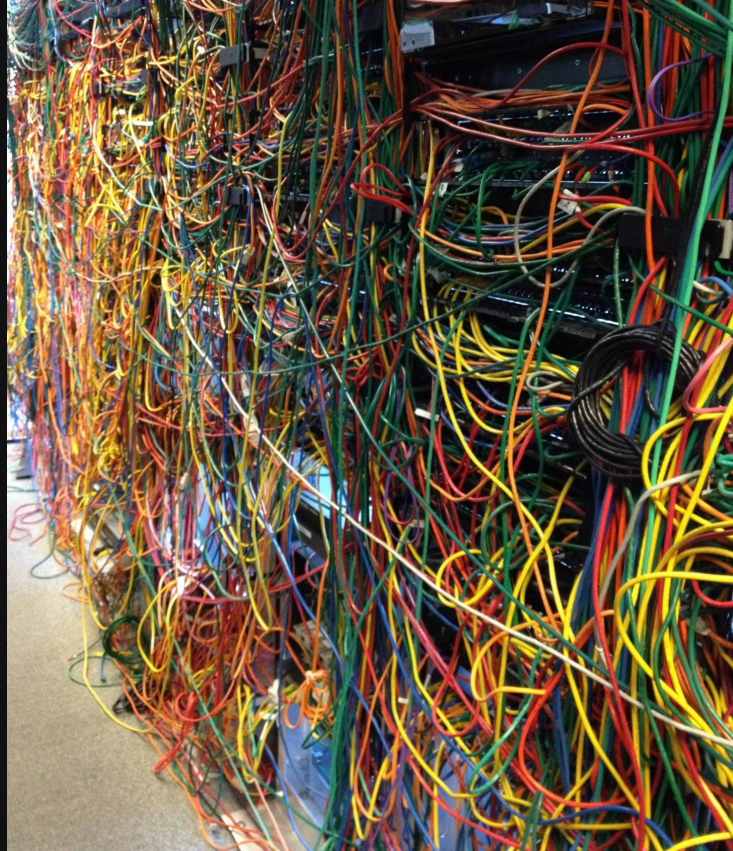
Agenda

- **Small intro to Docker**
 - Why?
 - Comparison to virtual machines
- **JBoss and Docker**
 - Current state
- **Demos**
 - Basic demos
 - WildFly demos



docker

Applications are complex



http://www.reddit.com/r/pics/comments/nf8la/im_always_tempted_to_switch_two_random_cables_to/

Different deployment environments



<http://www.shedworking.co.uk/2012/03/nuts-and-bolts-cable-ties.html>



docker

A project to manage **containers**
...and **images**

Containers?

Yes, a lightweight operating system virtualization.

Virtual machines?

No, it's more lightweight.

VM (KVM, VMware)

We need to run the **whole OS** to start the process.

Container (LXC, Docker)

We launch processes **directly**, without booting the OS.

So, how this compares to a virtual machine?

It's **completely** different.

VM vs. container

	Virtual Machine (KVM, VMware)	Container (LXC, Docker)
Hardware	Simulated	Uses it (almost) directly
Supported OS'es	Almost any	Only Linux
Startup time	Seconds to minutes	Miliseconds
Scalability	A few	Sky is the limit (thousands)
Size	HUGE	Small

Linux FTW!

<https://github.com/boot2docker/boot2docker>

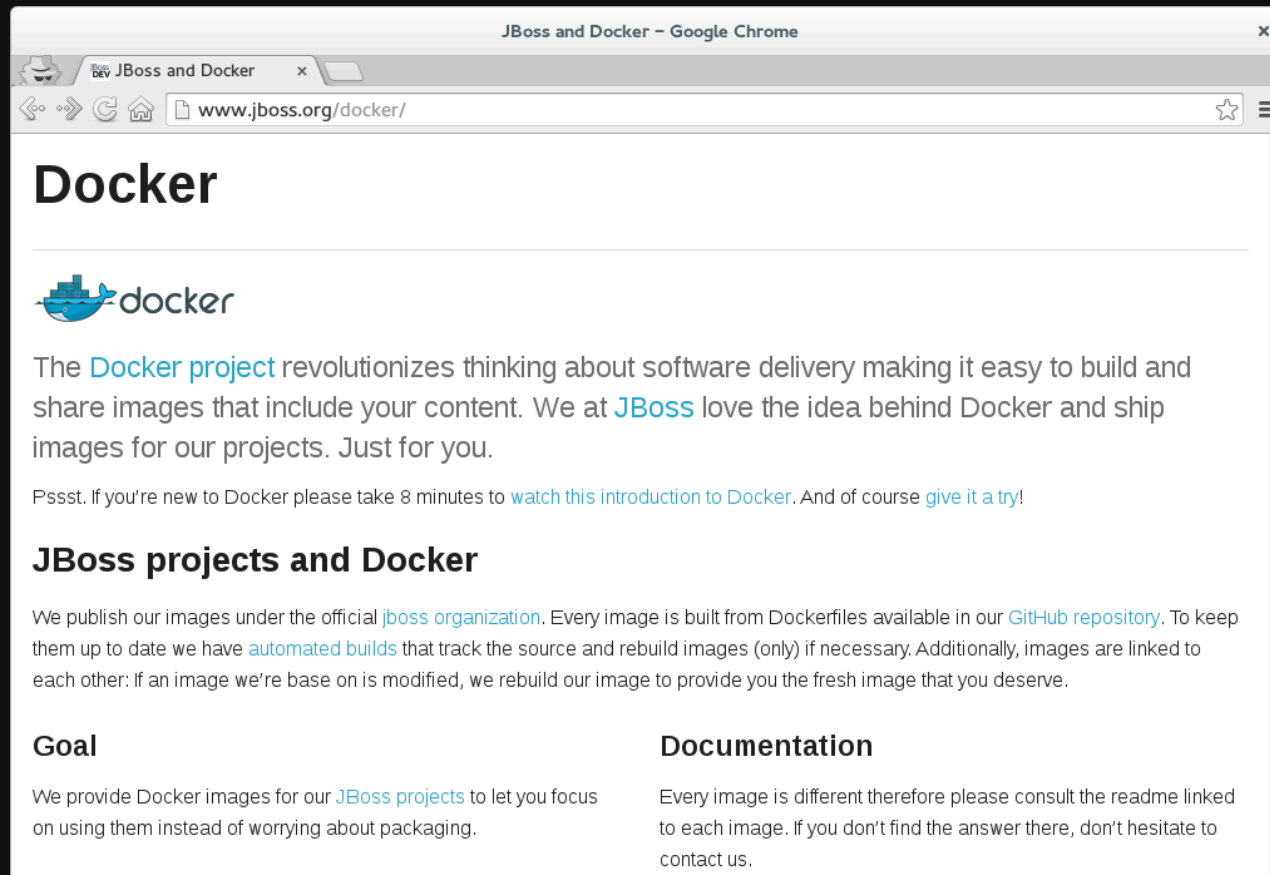
Docker community...

- **GitHub**
 - >1100 watchers
 - >14000 stars
 - >2700 forks
 - >660 open issues
 - >100 pull requests
- Over **585 contributors**
 - avg 17 commits / day
 - **90% are external**
- Many meetups all over the world
 - **50** active **meetup** groups

JBoss and Docker

Dockerized projects

- WildFly
- Keycloak
- TorqueBox
- Aerogear
- Immutant
- Nodyn
- LiveOak
- SwitchYard
- ...



<http://www.jboss.org/docker/>

View Public Profile | Docker Hub Registry - Repositories of Docker Images - Google Chrome

View Public Profile | x

https://registry.hub.docker.com/repos/jboss/

 What Is Docker? Use Cases Try It! Browse Install & Docs Log In Sign Up

jboss

Repositories

Show: All Sort by: Last Updated

Filter by name...

	jboss/keycloak-adapter-wildfly ✓ a minute ago	☆ 0	☁ 88
	jboss/keycloak-examples ✓ 5 days ago	☆ 0	☁ 20
	jboss/keycloak ✓ 5 days ago	☆ 0	☁ 117

<https://registry.hub.docker.com/repos/jboss/>

GitHub

54 commits 1 branch 0 releases 10 contributors

branch: master dockerfiles / +

Updated Keycloak to 1.0-rc-2

jpikrohlting authored 18 minutes ago latest commit a37863de12

aerogear	Removing java and maven installations.	2 months ago
immutant	Bump immutant to 1.1.4	5 days ago
keycloak	Updated Keycloak to 1.0-rc-2	18 minutes ago
liveoak	Update to LiveOak Alpha02	2 days ago
nodyn	Removing non required user and updating readme file	20 days ago
torquebox	TorqueBox upgrade to 3.1.1	2 months ago
wildfly	wildfly: Perform chown in same step as untar, to save 100Mb	16 days ago
LICENSE	Updated README + added LICENSE file	3 months ago

Code

Issues 6

Pull Requests 2

Pulse

Graphs

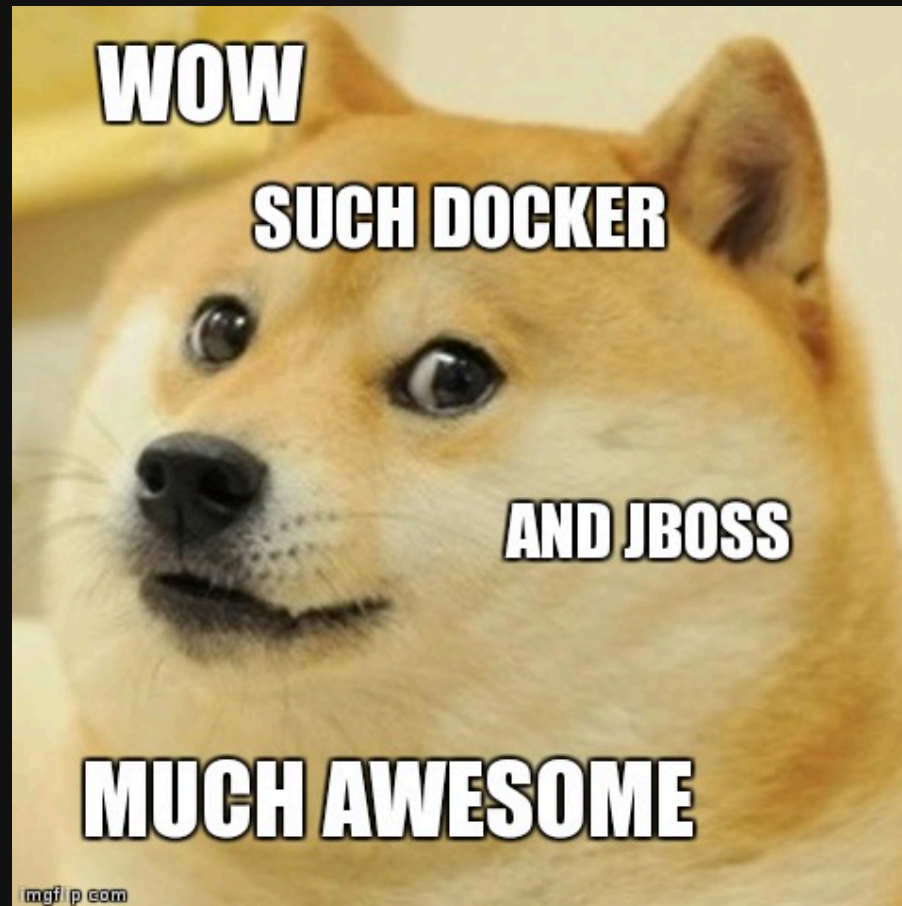
HTTPS clone URL

https://github.cc

You can clone with HTTPS or Subversion

Download ZIP

<https://github.com/jboss/dockerfiles>



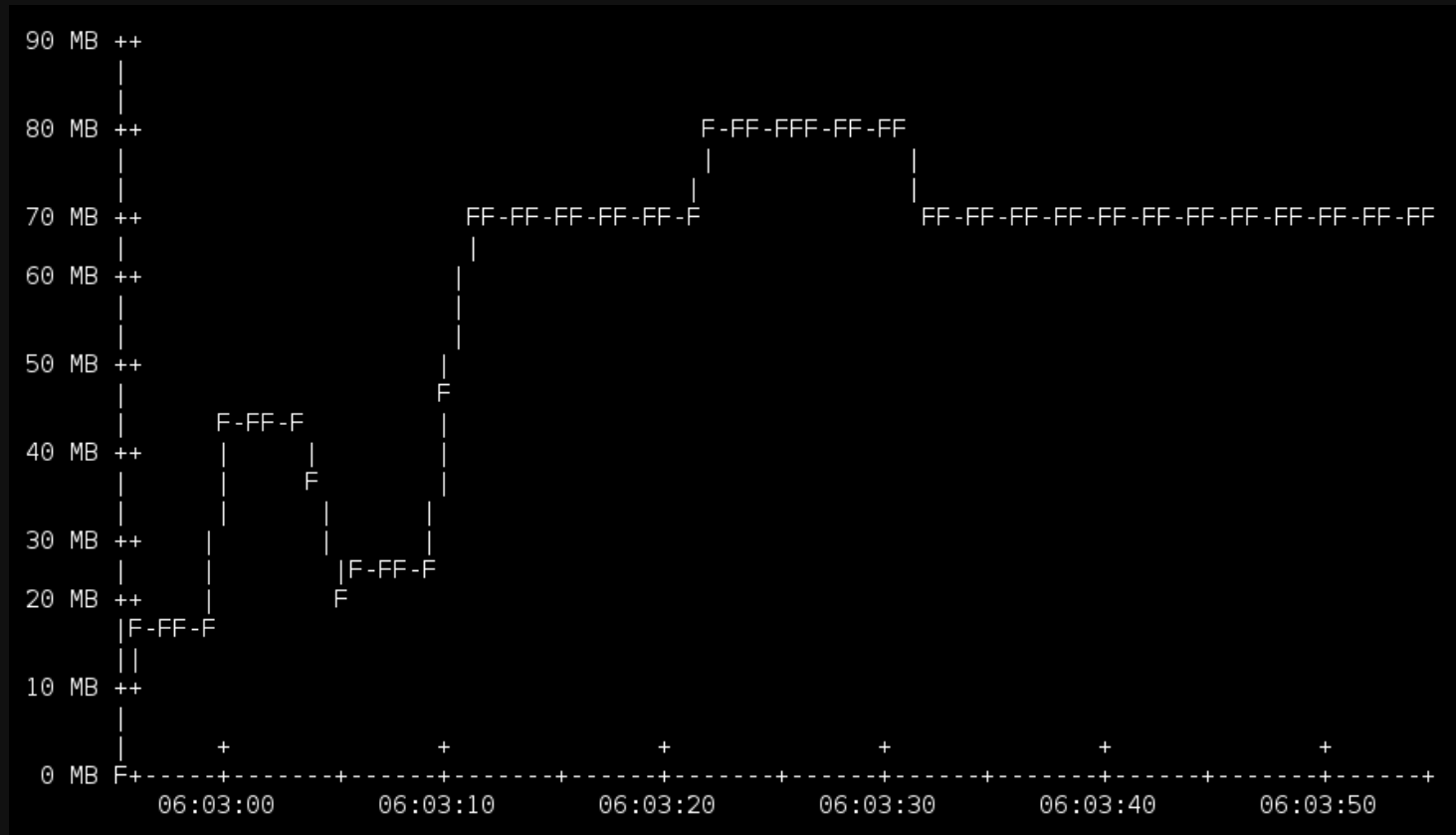
Demos!

A few tips

I need metrics!

- Use iptables for network
- Parse /sys/fs/cgroup/ for everything else

/sys/fs/cgroup/ + gnuplot



I need to limit resources!

<https://goldmann.pl/blog/2014/09/11/resource-management-in-docker/>

I need to do backup!

- Files?
 - Use volumes
- Data?
 - Use links

I need to access my containers!

- Do you own the host?
 - Use nsenter (soon docker exec)
- In other cases:
 - Run sshd inside of the container (boooo...)

I need performance!

RLY?

- CPU: native
- Memory: (almost) no overhead
- Network: no overhead with `--net host`
- I/O: native on volumes

Thanks!



Attributions:

<https://www.docker.io/>

